

CHORDATES

Only three to four per cent of known animal species are chordates, yet they include the largest, fastest, and most intelligent animals alive today. Most chordates have a skeleton made of bone or cartilage, but their defining feature is the rod-like notochord – the evolutionary forerunner of the spine.

Fossil records indicate that the earliest known true chordates were small, streamlined animals, just a few centimetres long. Living at least 500 million years ago, they had no hard parts, except for a stiff but flexible cartilaginous notochord. The notochord ran the length of their body, creating a framework for their muscles to pull against. Today's chordates all have this feature, and a small number keep it throughout life. However, in the vast majority of chordates – from fish and amphibians to reptiles, birds, and mammals – the notochord is present only in the early embryo. As the embryo develops, the notochord disappears, replaced by an internal skeleton made of cartilage or bone. These animals are known as vertebrates, from the column of vertebrae that makes up their spine.

Unlike shells or body cases, bony skeletons work on an amazing spectrum of scales. The smallest vertebrate, a freshwater fish called *Paedocypris progenetica*, is less than 1 cm ($\frac{3}{8}$ in) long and weighs several billion times less than a blue whale, which is the largest chordate, and the biggest animal that has ever lived.

LIFESTYLES

It is perhaps surprising that one of the simplest chordate groups, the sea squirts (tunicates), do not have a skeleton or backbone, and spend their adult lives fastened in one place. However, in their planktonic larval stage they do have a notochord, indicating their shared ancestry with vertebrate chordates such as us. In contrast to these simple chordates, vertebrate chordates are often fast-moving animals, with rapid reactions, a well-developed nervous system, and a sizeable brain. Birds, mammals, and a few fish use energy from food to keep their body at a constant optimum temperature.

Chordates vary in the way they reproduce and raise their young. All mammals bear live young, apart from the duck-billed platypus and echidnas. Almost all other chordates lay eggs, though there are examples of live-bearing species in all vertebrate groups except birds. The number of offspring produced is related to the amount of parental care. Some fish produce millions of eggs, and play no part in raising their young, whereas mammals and birds have much smaller families.



TUNICATES

The subphylum Tunicata contains just over 3,000 species. Larval tunicates have a notochord, and resemble tadpoles. The adults are filter feeders, the most common type being sea squirts.

AMPHIBIANS

Numbering over 8,200 species, amphibians include frogs and toads, salamanders, newts, and caecilians. Most start their lives as aquatic larvae, but as adults they take up life on land.



CHORDATE TREE

